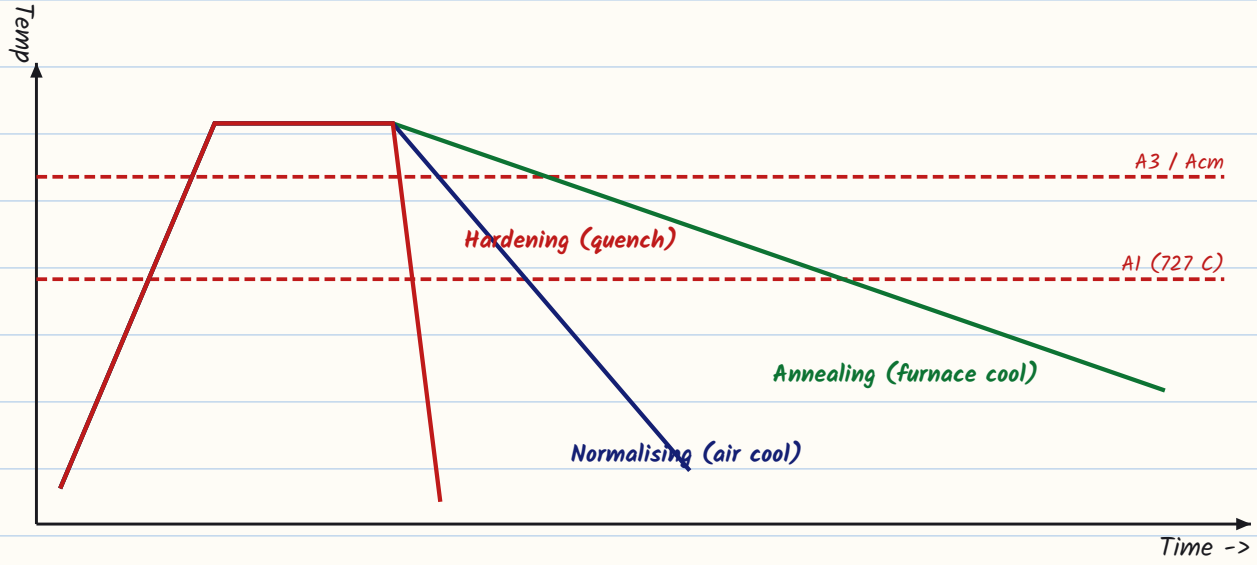
UNIT - III : HEAT TREATMENT OF STEELS

1. Introduction

* HEAT TREATMENT = controlled heating and cooling of a metal in solid state to obtain desired structure and properties (hardness, strength, toughness, ductility, machinability, stress relief).
* Any heat treatment has 3 steps: (i) HEATING to a definite temperature at a suitable rate, (ii) SOAKING (holding) for sufficient time for uniform temperature and structure change, (iii) COOLING at a definite rate (furnace, air, oil, water, brine).
* Soaking time: about 1 hour per 25 mm thickness (+ heat up time). Heating atmosphere: oxidising / reducing / neutral / vacuum / salt bath to avoid scaling and decarburisation.



2. Effect of Alloying Elements on Fe - Fe3C System

* Alloying elements shift the eutectoid TEMPERATURE and eutectoid COMPOSITION (% C) and change the austenite phase field.

* AUSTENITE STABILISERS (expand gamma field, lower A1 and A3): Ni, Mn, Co, Cu, C, N. With enough Ni (> 8%) / Mn (12%) austenite is retained at room temperature (austenitic stainless steel, Hadfield steel).
* FERRITE STABILISERS (shrink gamma loop, raise A1): Cr, Si, Mo, W, V, Al, Ti, Nb. With > 13% Cr the gamma field vanishes completely - ferritic stainless steels.
* Eutectoid carbon content is LOWERED by almost all alloying elements (Ti, Mo, W, Si, Cr strongly): e.g. at 5% Cr, eutectoid is at about 0.5% C; the eutectoid temperature is raised by Cr, W, Mo, Si and lowered by Ni, Mn.

Element	A1 temp	Eutectoid % C	Type
Ni	Lowers	Lowers	Austenite former
Mn	Lowers	Lowers	Austenite former
Cr	Raises	Lowers	Ferrite former + carbide
Mo, W, V	Raises	Lowers	Ferrite former + strong carbide
Si	Raises	Lowers	Ferrite former, graphitiser

* Carbide formers (Cr, W, Mo, V, Ti) produce alloy carbides (Cr_7C_3, WC, Mo_2C, VC) -> hardness, wear resistance and secondary hardening; most alloying elements (except Co) SHIFT THE TTT CURVE TO THE RIGHT (delayed transformation) -> improved hardenability.

3. Annealing

* Heating steel above (or around) critical temperature, soaking and cooling SLOWLY (in furnace) to obtain a soft, ductile, stress free,

coarse pearlite structure.

* Purposes: soften steel for machining or cold working, relieve internal stresses, refine grain, remove gases, improve ductility, homogenise structure, prepare for further heat treatment.

Types of annealing

* FULL ANNEALING: hypoeutectoid steel heated to 30 - 50 C ABOVE A3 (hypereutectoid: above A1), soaked, cooled in furnace slowly (10-30 C/h). Structure: coarse pearlite + ferrite (or cementite).
* PROCESS (sub-critical) ANNEALING / recrystallisation anneal: heating cold worked low carbon steel to 550-650 C (below A1), restores ductility by recrystallisation.
* STRESS RELIEF annealing: 500 - 650 C for welded / cast / machined parts, slow cool. No structure change.
* SPHEROIDISING annealing: high carbon steel (> 0.6% C) held just below A1 (650-700 C) for long hours (or cycled about A1) -> cementite lamellae turn into SPHEROIDS in ferrite; best machinability, maximum softness (spheroidite).
* DIFFUSION / HOMOGENISING annealing: 1000 - 1200 C for long time to remove coring and segregation in ingots and castings.

4. Normalizing

* Heating 30 - 50 C ABOVE A3 (for hypoeutectoid) or above Acm (hypereutectoid), soaking, then cooling in STILL AIR.
* Faster cooling than annealing gives FINE pearlite with finer grains, more strength and hardness (about 15-20% more than annealed), good toughness, uniform structure.

Annealing	Normalising
Furnace cooling (slow)	Air cooling (faster)
Coarse pearlite; softest	Fine pearlite; harder, stronger
Costly, time consuming	Cheaper, quicker
For better machinability, cold working	To refine grain, remove casting / forging stresses, improve strength
Lower strength, high ductility	Higher strength, good toughness

* Used for castings, forgings, welded parts and before hardening to get a uniform structure.

5. Hardening

* Heat to 30 - 50 C above A3 (hypoeutectoid) or above A1 (hypereutectoid, NOT above Acm), soak till all pearlite changes to austenite, then QUENCH rapidly (faster than critical cooling rate) in water, brine, oil, or polymer to get MARTENSITE.
* MARTENSITE: super-saturated interstitial solid solution of carbon in BCT (body centred tetragonal) iron, needle / lath shaped; formed by diffusionless shear transformation of austenite between Ms and Mf temperatures; very hard (up to 65 HRC), strong and brittle.

Austenite (FCC) --rapid quench--> Martensite (BCT)

* Hardness of martensite depends on CARBON content only (0.6% C gives about 65 HRC). Ms (C) = 539 - 423 (%C) - 30.4 (%Mn) - 17.7 (%Ni) - 12.1 (%Cr) - 7.5 (%Mo) degrees C. Above 0.5% C a part of austenite remains as RETAINED AUSTENITE.
* Quenching media in order of severity: Brine > Water > Mineral oil >

Polymer > Air > Furnace. Quench stages: vapour blanket -> boiling -> convection.

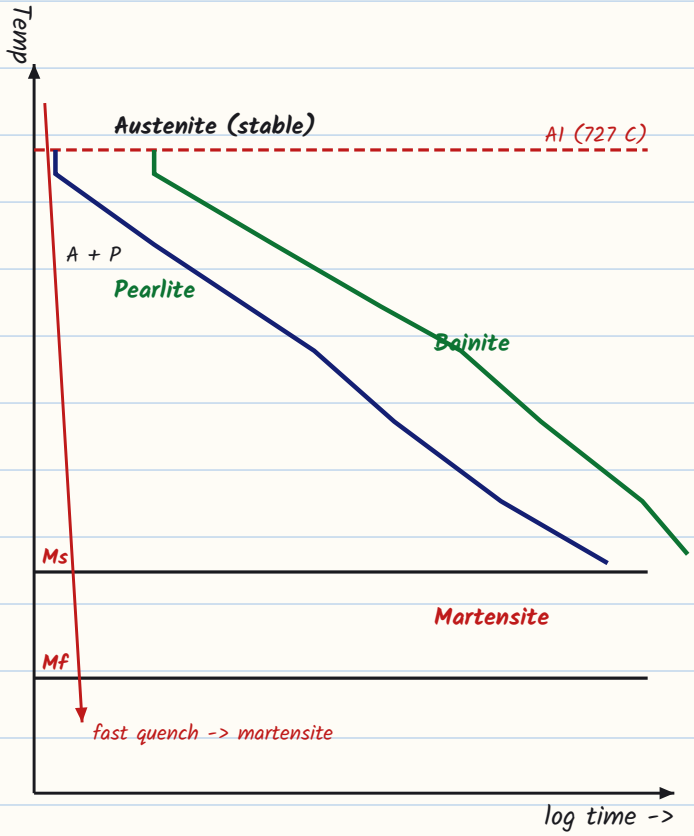
* Defects: quench cracks, distortion, soft spots, residual stresses, decarburisation. Remedies: use alloy steel with oil quench, interrupted quench, tempering immediately.

Critical cooling rate (CCR)

* The minimum cooling rate which just avoids the pearlite / bainite 'nose' of the TTT curve and gives fully martensitic structure. Alloying (except Co) lowers CCR -> easier to harden.

6. TTT Diagrams (Isothermal Transformation)

* TTT = Time - Temperature - Transformation (also IT or C-curve or S-curve) diagram: shows time needed for austenite to transform isothermally into other phases at a constant temperature below A1. Obtained from many small samples quenched into salt baths at different temperatures, held for various times, and quenched; microstructure / hardness / dilatometry used to find fraction transformed.



* LEFT curve = start of transformation (about 1% transformed), RIGHT curve = finish (99%). Left of start curve: only unstable austenite. Between: austenite + product. Right of finish: all product.
* Eutectoid steel: 727 - 550 C: PEARLITE (coarse at high T, fine at lower T; hardness 15 - 40 HRC). Nose at about 550 C (minimum incubation time about 1 s). 550 - 250 C: BAINITE (upper bainite - feathery, lower bainite - needle like; 40 - 55 HRC; tough). Below Ms (~ 220 C for eutectoid steel): MARTENSITE (athermal; depends only on temperature, not time).
* Cooling curves superposed on TTT diagram: slow cooling (annealing) -> coarse pearlite; moderate (normalising) -> fine pearlite; faster that misses the nose (> CCR) -> martensite; interrupted quenching -> bainite.
* Hypo / hypereutectoid steels have an extra curve (proeutectoid ferrite / cementite) above the pearlite nose. TTT is valid ONLY for isothermal

hold; for continuous cooling use CCT diagram (curves shifted to right and down).

Variations in hardening using TTT

* MARTEMPERING (marquenching): quench austenite to just above Ms in hot oil / salt, hold till temperature equalises, then air cool through the Ms-Mf range -> martensite with less distortion, cracking. Then temper.
* AUSTEMPERING: quench to 250 - 400 C (above Ms), hold isothermally until all austenite becomes BAINITE, cool in air. Gives good toughness and ductility with high hardness, no quench cracks, no tempering needed (springs, small tools).

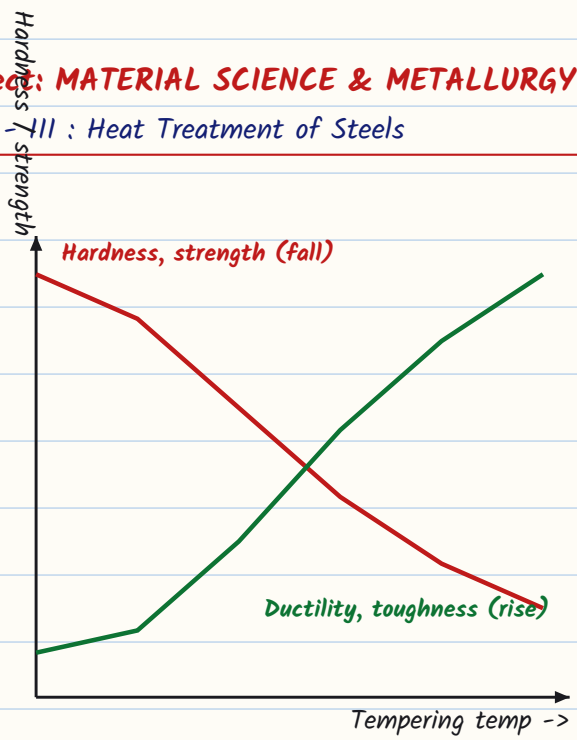
7. Tempering

* Reheating hardened (martensitic) steel to a temperature BELOW A1 (150 - 650 C), holding, and cooling (usually air). It reduces brittleness and internal stresses and improves toughness and ductility at the cost of some hardness / strength.

Temp (C)	Stage / structure	Properties and use
150 - 250	1st stage: epsilon carbide, tempered martensite	Hard (60 HRC), slight toughness - cutting tools, files, razors
250 - 350	2nd stage: retained austenite decomposes to bainite	Spring steel, chisels (400-450 for springs)
350 - 450	3rd stage: cementite precipitates, troostite	Hardness 40-50 HRC, good toughness - springs, hammers
450 - 650	Cementite coarsens, sorbite; spheroidised	Max toughness, 25-35 HRC - gears, shafts, bolts, axles (QUENCHED + TEMPERED = 'hardened and tempered', 'bonnet')

* TEMPER EMBRITTLEMENT: loss of toughness on tempering at 250 - 400 C ('blue brittleness' / 300 C embrittlement) and on slow cooling through 450 - 600 C (impurities P, Sb, Sn segregate to grain boundaries; avoided by adding Mo and by quick cooling).
* SECONDARY HARDENING: Cr, Mo, V, W steels harden again at 500-600 C due to alloy carbide precipitation (red hardness of HSS).

Hardening + Tempering = Quenching and tempering (Q & T) - the most important heat treatment for strong, tough steel parts.



Microconstituents of steel – quick table

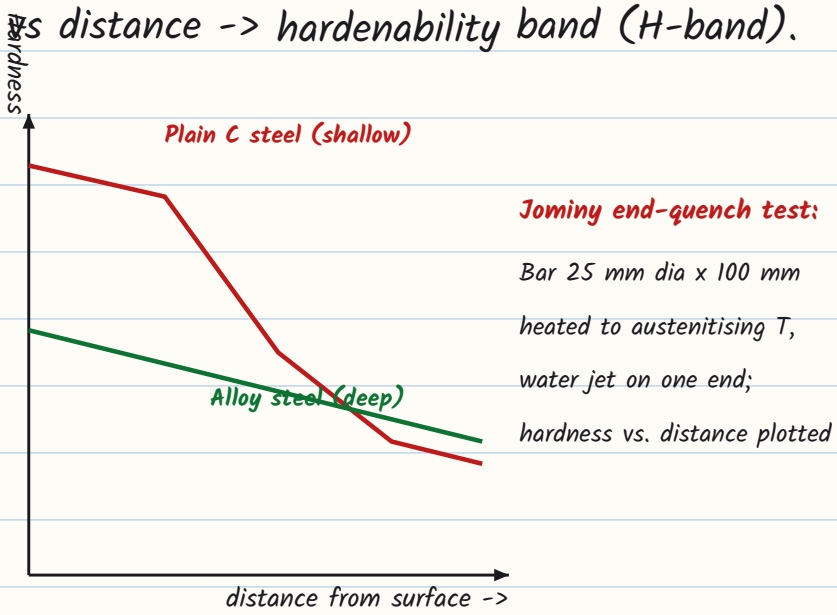
Constituent	Formation	Structure	Hardness (approx.)
Ferrite	Slow cooling	BCC, soft	80-100 HB
Pearlite (coarse / fine)	Eutectoid transformation 727-550 C	Ferrite + Fe3C lamellae	200-300 HB
Spheroidite	Long hold below A1	Cementite spheroids	150-200 HB (soft)
Bainite	Isothermal 550-250 C	Ferrite + fine carbide	40-55 HRC
Martensite	Quench below Ms	BCT supersaturated	55-67 HRC
Troostite (tempered 400)	Tempering	Fine carbide in ferrite	40-50 HRC
Sorbite (tempered 600)	Tempering	Coarser carbide	25-35 HRC

* Retained austenite (Mf below room temp. for > 0.5% C steels) is soft, unstable, causes dimensional change with time (aging) - removed by

sub-zero treatment or double tempering.

8. Hardenability

* Hardenability = ability of steel to be hardened to a depth by quenching (depth of martensite formation); NOT the maximum hardness (hardness depends on %C). It is governed by the position of the TTT nose - alloys shift it right.
* JOMINY END-QUENCH TEST (ASTM A255): standard bar 25 mm dia. x 100 mm long, austenitised, placed in fixture and sprayed with water jet at one end. After quenching, 2 flats are ground and HRC hardness measured at every 1.6 mm distance from quenched end. Plot hardness vs distance -> hardenability band (H-band).



* Distance at which hardness falls to 50% martensite hardness = 'ideal critical diameter' D_I. Alloy steels (Cr, Mo, Ni, Mn) show flat curves = high hardenability; plain carbon steel falls steeply.
* Factors: carbon content, alloy elements (except Co), austenitic grain size (coarse = higher), homogeneity of austenite, quenching medium (severity H value), section size (mass effect).

Quenching media and severity

Medium	Severity H	Used for	Remarks
Brine (10% NaCl)	2.0 - 5.0	Plain carbon steel, simple shapes	Very fast, high risk of cracks
Water	1.0	Low alloy and carbon steels	Fast, distortion risk
Oil (mineral)	0.25-0.7	Alloy steels, tools	Slower, less distortion, fire risk
Polymer solution	0.3-1.0	Induction / large parts	Adjustable severity
Air / gas	0.02	High speed steels, high alloy	Very slow, minimum distortion
Molten salt / lead	Variable	Austempering, martempering	Uniform, no vapour film

* Hot working (hardening) furnaces: box, pit, salt bath, vacuum, fluidised bed, induction. Protective atmospheres: endothermic gas (CO, H2, N2), nitrogen based, to prevent scaling and DECARBURISATION.
* Typical heat treatment of a gear: normalise forging (to refine grain) -> machine -> carburise -> quench in oil -> sub-zero -> temper at 180 C -> finish grind.

9. Surface Hardening Methods

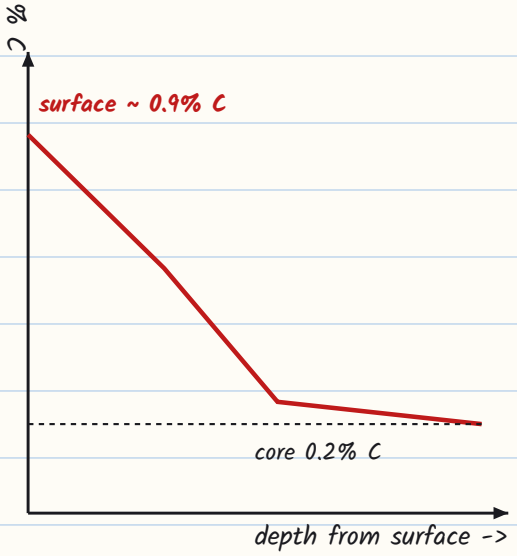
Aim: HARD wear resistant case (surface) with TOUGH shock absorbing core. Two ways: (A) change surface composition (case hardening), (B) change only structure by selective heating and quenching.

A. Carburising (case hardening) - low carbon steel (0.1-0.25% C)

* Pack carburising: parts packed in iron box with charcoal + 10-20% BaCO3 energiser, 900 - 950 C for 6 - 12 h; case depth 0.5 - 1.5 mm.

Gas carburising: furnace atmosphere of CH4 / endo-gas (CO + H2), better control; Liquid carburising: cyanide salt bath (NaCN) 850-900 C, case depth 0.1-0.5 mm (shorter time).

* After carburising: parts are quenched (direct or after re-heating) and tempered at 150 - 200 C. Surface: high carbon martensite (0.8-1% C, 60+ HRC); core: low carbon ferrite-pearlite tough. Uses: gears, camshafts, piston pins, chain links.



B. Nitriding - alloy steel containing Al, Cr, Mo, V (Nitralloy)

* Steel (already hardened and tempered) heated at 500 - 550 C in ammonia (NH3) gas for 20 - 100 h; nascent nitrogen forms very hard nitrides (AlN, CrN, Fe4N). NO quenching, so little distortion.
* Case depth 0.2 - 0.7 mm, hardness 900 - 1100 HV (above carburising), good fatigue, wear and corrosion resistance, keeps hardness up to 500 C. Uses: crankshafts, cylinder liners, spindles, gears, dies. Ion (plasma) nitriding is a modern process.

C. Cyaniding and carbonitriding

* CYANIDING: liquid bath of NaCN + NaCl + Na2CO3 at 800 - 870 C; steel absorbs both C and N; case 0.1 - 0.3 mm in 30 min; quench in oil / water. Used for small parts - screws, nuts, bolts. Toxic bath.

* CARBONITRIDING: gas process with NH3 + hydrocarbon at 750 - 850 C; safe alternative to cyaniding.

D. Flame hardening

* Surface of medium carbon steel (0.4 - 0.6% C) heated by oxy-acetylene flame above A3 and immediately quenched by water spray; case 2-6 mm; no change in surface composition. Good for large parts, gear teeth, lathe beds, rails; parts then tempered.

E. Induction hardening

* Part placed in copper coil carrying high frequency current (10 - 500 kHz) -> eddy current heats the skin rapidly (seconds); quench by water spray. Depth controlled by frequency (higher f -> thinner case). Fast, clean, automatic, repeatable. Uses: crankshaft journals, gears, axles, camshafts, valve seats.

Process	Steel	Temp (C)	Case depth	Hardness
Carburising	0.1-0.25% C	900-950	0.5-1.5 mm	58-62 HRC
Nitriding	Al, Cr, Mo alloy	500-550	0.2-0.7 mm	900-1100 HV
Cyaniding	Low C	800-870	0.1-0.3 mm	55-60 HRC
Flame	0.4-0.6% C	> A3	2-6 mm	50-60 HRC
Induction	0.4-0.6% C	> A3	1-5 mm	50-60 HRC

10. Age (Precipitation) Hardening

* Used for NON-ferrous alloys (Al-Cu duralumin, Al-Mg-Zn, Cu-Be, Ni super alloys, Mg alloys) whose solid solubility DECREASES with falling temperature (solvus line). Discovered by Wilm (1906) in duralumin.
* STEP 1 - SOLUTION TREATMENT: heat to single phase alpha region (e.g. Al-4% Cu at 500 C), soak to dissolve all CuAl2 in alpha.

* STEP 2 - QUENCHING in water: gives supersaturated solid solution (SSSS), soft and unstable.
* STEP 3 - AGEING: natural (room temperature, 4-5 days for duralumin) or artificial (100 - 200 C for several hours). Fine coherent precipitates (GP zones -> theta" -> theta' -> equilibrium theta CuAl2) form; strain fields block dislocations -> hardness and strength increase.
* OVERAGEING: long time / high temperature -> precipitates coarsen, coherency lost, hardness drops. Peak hardness at optimum time.

SSSS --> GP zones --> theta" --> theta' --> theta (CuAl2)

* Mechanism: precipitate particles obstruct dislocation motion (cutting or Orowan bowing). Maraging steels and PH stainless steels also age harden.

II. Cryogenic Treatment

* Deep cooling of material to very low temperature (-80 C shallow, -196 C in liquid nitrogen) slowly, holding 24 hours, and slow warming, usually followed by tempering. An extension of the hardening process.
* Effects: converts RETAINED AUSTENITE to martensite, precipitates fine eta-carbides (increase wear resistance), relieves residual stress, improves dimensional stability. Increases wear resistance 100 - 600%, hardness 1-3 HRC.
* Uses: cutting tools (HSS drills, milling cutters), dies, gauges, bearings, brake discs, gun barrels, engine parts, musical instruments. Never use rapid cooling (cracks).

Important questions

* 1. Effect of alloying elements on Fe-Fe3C diagram (austenite / ferrite stabilisers, eutectoid shift).
* 2. Annealing vs normalising: types, temperature range, structure and applications.
* 3. Draw a TTT diagram for eutectoid steel; mark Ms, Mf; show cooling curves for annealing, normalising, hardening, austempering, martempering.
* 4. Define hardenability; explain the Jominy end quench test.
* 5. Compare carburising, nitriding, cyaniding, flame and induction hardening.
* 6. Explain age hardening of Al-Cu alloy; what is overageing?

Revision: Annealing - soft; Normalising - fine and uniform; Hardening - martensite; Tempering - toughness; Case hardening - hard skin, tough core; Age hardening - non-ferrous, precipitate; Cryogenic - retained austenite removal.